IGCSE · Cambridge (CIE) · Biology

43 mins 14 questions

Target Test

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Total Marks **/43**

1 What are the smaller basic units of starch and glycogen molecules?

	starch	glycogen
A	amino acids	fatty acids and glycero
B	amino acids	glucose
C	glucose	fatty acids and glycero
D	glucose	glucose

(1 mark)

2 (a) A student processed a sample of food into an aqueous solution and tested it with a test solution. The test solution changed from blue to violet.

Name the solution used in this test.

.... **(1 mark)**

(b) State the food group that the experiment in part (a) showed a positive result for.

.... **(1 mark)**

(c) A sample of butter was shaken with ethanol, followed by an equal volume of cold water.

Predict the appearance of the mixture after the addition of water.

.... **(1 mark)**

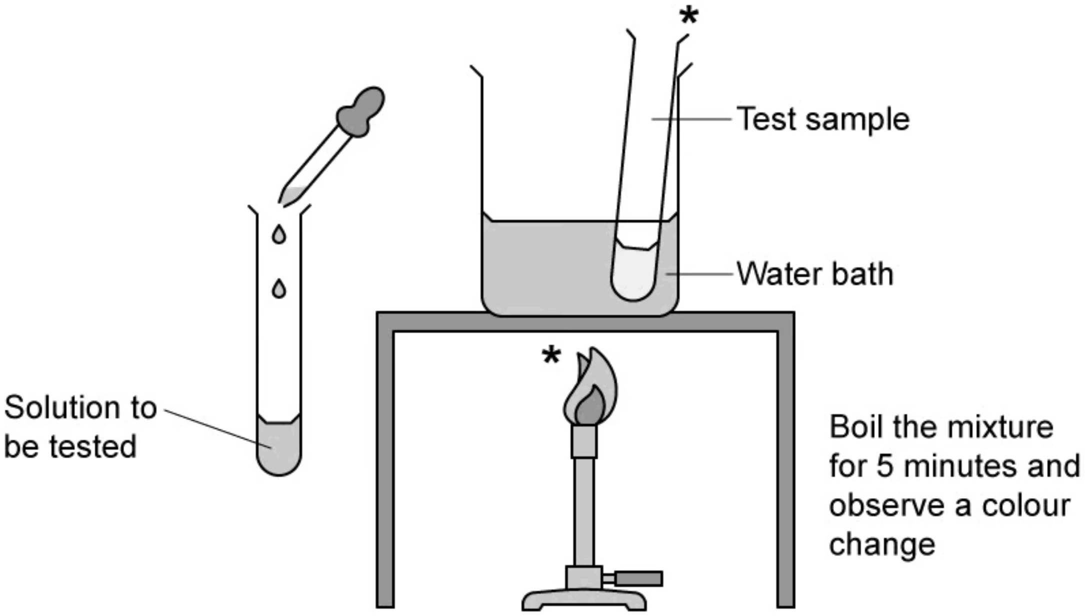
(d) A sample of freshly-squeezed orange juice was mixed with DCPIP solution.

Predict the result of this food test, and explain your prediction.

....

(2 marks)

3 The drawing shows a food test being carried out in a laboratory.



Which food test is being shown and which food group is being tested for?

A. Biuret test for proteins

B. DCPIP test for Vitamin C

C. Benedict's test for reducing sugars

D. Emulsion test for fats

(1 mark)

4 Extended only

DNA is a large molecule made from two chains of nucleotides held together by cross-links between pairs of bases.

Which of the following is a correct base pair?

A. T with C

B. G with A

C. C with G

D. C with A

(1 mark)

5 Which row shows the three main elements from the Periodic Table that are present in most molecules in living organisms?

A	carbon	oxygen	nitrogen
B	hydrogen	carbon	oxygen
C	oxygen	sulfur	carbon
D	hydrogen	carbon	nitrogen

(1 mark)

6 (a) Extended only

Fig. 1 shows a nucleotide.

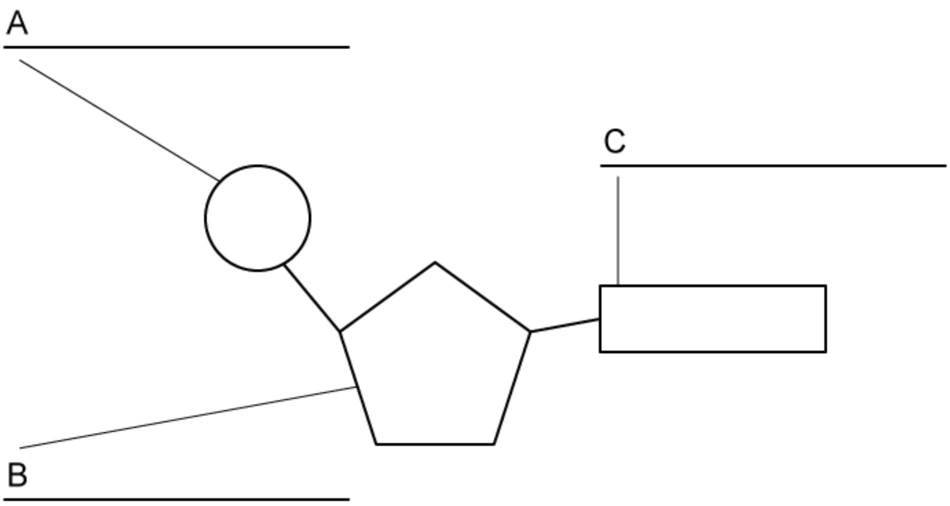


Fig.1

Label the parts of the nucleotide molecule shown as **A**, **B**, and **C**.

(3 marks)

(b) **Extended only**

State the base pairings between A, C, G and T in a DNA molecule.

.... **(1 mark)**

(c) Define the term 'double helix' in the structure of DNA.

....

.... **(2 marks)**

(d) State a part of the nucleotide shown in part (a) that occupies the outside of the DNA double helix.

.... **(1 mark)**

7 **Extended only**

Which row represents the three main constituents of a **DNA nucleotide**?

A	bases	phosphate	nucleus
B	deoxyribose sugar	nucleus	chromosome
C	bases	deoxyribose sugar	phosphate
D	phosphate	chromosome	bases

(1 mark)

8 A chemical test is carried out on a flax seed as follows:

1. A small sample of seed is crushed and placed in a test tube
2. Approximately 2 cm^3 of ethanol is added to the test tube containing the crushed seed
3. The mixture is allowed to settle for about 3 minutes and poured into a second test tube containing water
4. A milky/cloudy layer forms at the top of the test tube

Which of the following correctly names the test and the molecule that it tests for.

A. Biuret test for proteins

B. iodine solution test for starch

C. ethanol test for oils

D. emulsion test for lipids

(1 mark)

9 Which statement is false?

A. Fatty acids are a type of lipid

B. Glucose is a monosaccharide

C. A triglyceride consists of three fatty acids and one glycerol molecule

D. Amino acids each have a side group (R) that distinguishes them from other amino acids

(1 mark)

10 Which statement about the structure of DNA is correct?

A. Base A always pairs with base C.

B. Base A always pairs with base T

C. DNA is made of protein.

D. DNA forms a single helix.

(1 mark)

11 (a) **Fig. 1** shows five food tests and the food groups that those tests are designed to be used for.

Test	Food group being tested
Benedict's	Starch
Iodine	Protein
Ethanol emulsion	Vitamin C
DCPIP	Glucose
Biuret	Lipids

Fig. 1

Using five straight lines, link each food test to the food group that the test is used for.

(5 marks)

(b) Describe the colour change that is observed when a sample of food has tested positive in the Benedict's test.

....

.... **(2 marks)**

(c) (i)

Describe **two** safety precautions that need to be taken when performing the Benedict's test on a sample of food.

[2]

(ii)

Describe one **additional** safety precaution when performing the ethanol emulsion test on a sample of food.

[1]

....

....

.... **(3 marks)**

(d) A student tested a sample of three different foods for starch.

The student's partial results are shown in **Table 1**.

Table 1

food sample	colour of test sample after testing
lasagne	blue/black
fish	
bread	

Complete **Table 1** to predict the results that would be obtained.

....

.... **(2 marks)**

12 Which element is found in proteins but **not** carbohydrates?

A. carbon

B. hydrogen

C. nitrogen

D. oxygen

(1 mark)

13 List the chemical elements present in fats.

.... **(1 mark)**

14 (a) **Table 1** lists a number of chemical substances.

Table 1

hydrogen	
water	
sulfur	
silicon	
titanium	
nitrogen	
carbon	

Place a tick (✓) in the boxes next to the chemical elements that are found naturally in living organisms.

(3 marks)

(b) Name a group of biological molecules that contain **carbon**, **hydrogen** and **oxygen** only.

.... **(1 mark)**

(c) **Table 2** lists some carbohydrates.

Place a tick (✓) in the boxes that classify each carbohydrate as either a monosaccharide, a disaccharide or a polysaccharide.

Table 2

Carbohydrate	Monosaccharide	Disaccharide	Polysaccharide
Glycogen			
Glucose			
Starch			
Maltose			

....

....

....

.... **(4 marks)**

(d) Name the molecule that, in certain fats, binds to three fatty acid molecules.

.... **(1 mark)**